(ii) Let us prove that $|x| > r$ if and only if $x < -r$ or $x > r$.
Consider $|x| > r$. If $r < 0$, then every $x \in \mathbb{R}$ satisfies the inequality.
For $r \geq 0$, there are two possibilities to consider.

Case (1). If $x \geq 0$, then $|x| = x > r$.
Case (2). If $x < 0$, then $|x| = -x > r$, that is, $x < -r$.

So we have $|x| > r$, if and only if $x < -r$ or $x > r$, that is, $x \in (-\infty, -r) \cup (r, \infty)$.

Remark:

(i) For any $a \in \mathbb{R}$, $|x - a| \leq r$ if and only if $-r \leq x - a \leq r$ if and only if $x \in [a - r, a + r]$.
(ii) For any $a \in \mathbb{R}$, $|x - a| \geq r$ is equivalent to $x - a \leq -r$ or $x - a \geq r$ if and only if $x \in (-\infty, a - r] \cup [a + r, \infty)$.

Example 2.4 Solve $|x - 9| < 2$ for x.

Solution:
$|x - 9| < 2$ implies $-2 < x - 9 < 2$. Thus, $7 < x < 11$.

Example 2.5 Solve $\left|\dfrac{2}{x-4}\right| > 1$, $x \neq 4$.

Solution:
From the given inequality, we have that $2 > |x - 4|$.
That is, $-2 < x - 4 < 2$ and $x \neq 4$.
Adding 4 throughout the inequality, we obtain $2 < x < 6$ and $x \neq 4$.
So the solution set is $(2, 4) \cup (4, 6)$.



Exercise - 2.2

1. Solve for x:
   (i) $|3 - x| < 7$. (ii) $|4x - 5| \geq -2$. (iii) $\left|3 - \frac{3}{4}x\right| \leq \frac{1}{4}$.
   (iv) $|x| - 10 < -3$.
2. Solve $\frac{1}{|2x-1|} < 6$ and express the solution using the interval notation.
3. Solve $-3|x| + 5 \leq -2$ and graph the solution set in a number line.
4. Solve $2|x + 1| - 6 \leq 7$ and graph the solution set in a number line.
5. Solve $\frac{1}{5}|10x - 2| < 1$.
6. Solve $|5x - 12| < -2$.

2.4 Linear Inequalities

Recall that a function of the form $f(x) = ax + b$, $a, b \in \mathbb{R}$ are constants, is called a linear function, because its graph is a straight line. Here a is the slope of the line and b is the y-intercept. If $a \neq 0$, then x-intercept $x = \frac{-b}{a}$ is obtained by solving $f(x) = ax + b = 0$.
But there are situations where we need to consider linear inequalities.
For example to describe a statement like " *A tower is not taller than fifty feet.*"
If x denotes the height of the tower in feet, then the above statement can be expressed as $x \leq 50$.